Art Tech
Media
Data
Experience

KEQING YANG
SELECTED WORKS 2023–2026

Information Art & Design
Tsinghua University

(2026)

Design Portfolio

ABOUT ME



HI, I'M YANG KEQING

Education Experience

2024,9 ~ Present

BFA Candidate in Art and Technology

Information Art & Design
Academy of Arts & Design, Tsinghua University

-CORE COURSES:Physical Computing, Materials, Techniques and Prototyping, Integrated Design for Tangible Interaction, Creative Coding for Graphics, Information Visualization, Interface Design

Software Skill

Figma · Blender · Python · Arduino · Touch Designer · Unreal Engine

Leadership & Campus Activities

2024 ~ Present
Head of Publicity and Education
Student Branch of Tsinghua University Red Cross Society

2026.1
Team Leader
Winter Social Practice Team to Anshan, Liaoning

2025.8
Publicity Group Leader
Volunteer Teaching Practice Team to Kashgar, Xinjiang

2025.7
Team Member
Rural Revitalization Practice Team to Xiangzhou, Guangxi

2025.7
Volunteer Teacher
Star Dream Classroom Volunteer Program
- Taught picture-book classes for children receiving medical treatment in Beijing.

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DOMESTIC COMPANION ROBOT | EMOTIONAL OUTBURSTS


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INTERACTION | EXHIBITION DESIGN | CULTURAL LEARNING


03 WOVEN DISCIPLINES
INFORMATION VISUALIZATION | DATA STORYTELLING



"What we learn is often not calmness, but endurance.

01 EMOTIONAL BUFFERING

CREATIVE ROBOT | 2026 / 3 ~ 2026 / 4 ■ Mi Haipeng ■ Individual Project

Emotional Buffer | Domestic Companion Robot | Emotional Outbursts

- A companion robot for everyday emotional support.
- Inspired by fetch games, it builds interaction through a loop of throw, retrieve, and return.
- It provides a lightweight alternative to one-way stress-relief tools by creating emotional comfort through sustained response.

https://youtu.be/g0rK1yOrDm8

BACKGROUND

It Started with a Moment

A brief moment of frustration became the starting point of this project.



After losing a competitive game,
I often felt an instant wave of frustration.

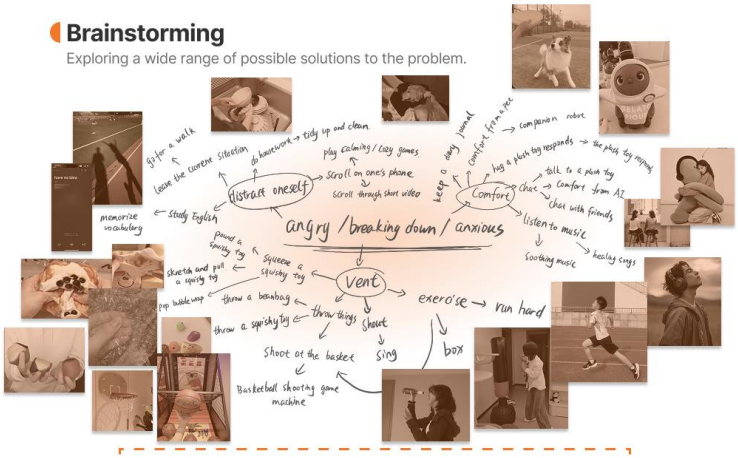
Sometimes, I wanted to throw something.
Not to destroy it, but to let the emotion out.

Yet everything around me felt wrong:
TOO FRAGILE, TOO EXPENSIVE,
TOO EMOTIONALLY LOADED.

This moment led me to think about emotional release as a design problem.

Brainstorming

Exploring a wide range of possible solutions to the problem.



Three Solution Directions

Distraction **# Emotional Comfort** **# Stress Relief**

RESEARCH

Theoretical Research & Literature Review

Bushman (2002) argued that intense **emotional venting** not only fails to reliably reduce anger, but may even intensify anger and aggression. Therefore, negative emotions cannot be effectively managed through venting alone. In contrast, Sheppes et al. (2011) found that under high-intensity negative emotions, individuals are more likely to choose lower-demand regulation strategies such as **distraction**. Consistent with this, Shafir (2016) argued that emotions are bidirectionally linked with **movement, posture, interoception, and proprioception**, suggesting that intentional changes in bodily movement and sensory input can help alter emotional experience. However, emotion regulation is not solely an internal process. Zaki and Williams (2013) proposed that people often rely on the **responses of others** to reshape their emotional states.

I aim to design a robot that uses subtle movements to redirect attention, helping users disengage from emotion and regulate it through light interaction and feedback.

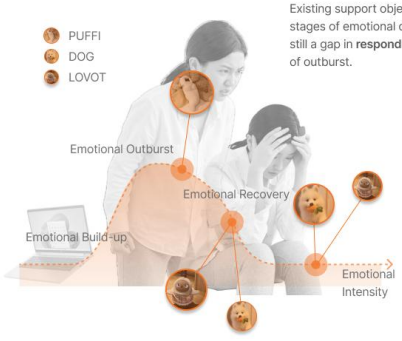
Bushman, B.J. (2002) 'Does venting anger feed or extinguish the flame? Catharsis, rumination, distraction, anger, and aggressive responding', Personality and Social Psychology Bulletin, 28(6), pp. 724–731.
Sheppes, G., Scheibe, S., Suri, G. and Gross, J.J. (2011) 'Emotion-regulation choice', Psychological Science, 22(11), pp. 1391–1396.
Shafir, T. (2016) 'Using movement to regulate emotion: Neurophysiological findings and their application in psychotherapy', Frontiers in Psychology, 7, article 1451.
Zaki, J. and Williams, W.C. (2013) 'Interpersonal emotion regulation', Emotion, 13(5), pp. 803–810.

Competitor Analysis



They all respond to high-arousal emotions through non-verbal and light interaction, helping users shift attention away from their feelings.

[How Different Supports Work Across the Emotional Curve]



Existing support objects cover different stages of emotional development, but there is still a gap in **responding** to the very moment of outburst.

[How Accessible Each Product Is Across Emotional Situations]

	DOG	LOVOT	PUFFI
Instantly Accessible	○	○	●
Depends on Active Response	●	●	○
Suitability for Emotional Peaks	●	●	○
Suitability for Emotional Peaks	●	○	○

Immediate release from PUFFI **+ emotional companionship** from pets / LOVOT
= a new form of emotional support

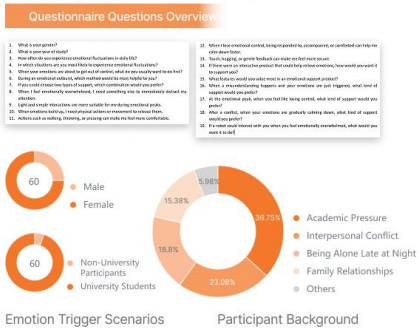
User Research

To better understand users' emotional support needs, I conducted a questionnaire survey focusing on **emotional fluctuation**, **coping behaviors**, and **expectations for an emotional support robot**.

Research Data Analysis

Of the **60** questionnaires distributed, **55** were valid.

Questionnaire Questions Overview



Emotion Trigger Scenarios

Participant Background

User Portrait

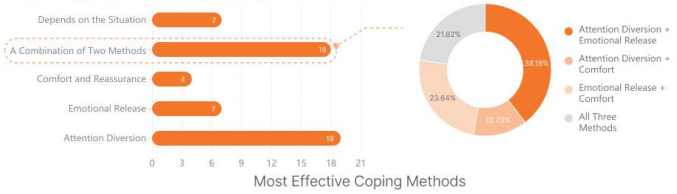


Name: Yang Yuan
Age: 20
Gender: Female
Location: Beijing
Identity: Sophomore Student

Emotional Pattern:
emotional stress / academic pressure / interpersonal tension / late-night solitude

Behavior & Need:
quiet / private / calm down / avoid talking immediately

Most Effective Coping Methods



Most Effective Coping Methods

Users need flexible emotional support for different situations, especially ways to distract themselves and release emotions.

Support Needs Across Emotional Stages





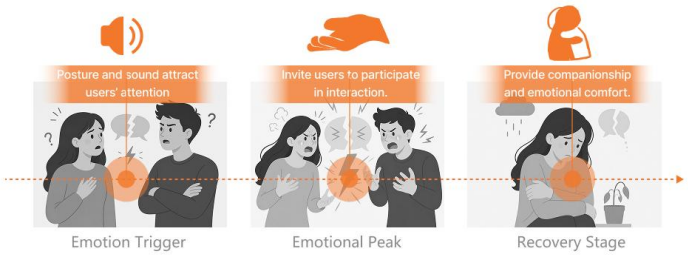
Users' support needs shift across emotional stages: attention diversion at the beginning, emotional release at the peak, and comfort during recovery.

Key Findings

Stage-Based Needs
Users' emotional needs change across different emotional stages.

Core Support Types
Attention diversion, emotional release, and comfort are the three main support needs.

Flexible Interaction
A single fixed response is not enough; the robot should provide flexible and stage-based interaction.



DESIGN DEFINITION



User Needs Identified from Research	Function Directions
Difficulty Shifting Attention	Attention Diversion
High-Arousal Emotions Need Immediate Release	Emotional Release
Verbal Communication Can Feel Burdensome	Non-Verbal Interaction
Emotional Needs Change Over Time	Stage-Based Interaction
Lack of Gentle Companionship	Comfort & Companionship

Design Positioning

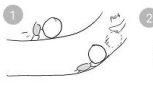
This project aims to create an emotional buffer zone: providing a gentler way of emotional expression for those with intense emotions.

Design Sketches

Initial sketches exploring Booji's form, proportions, and movement.

Interaction Mode Design

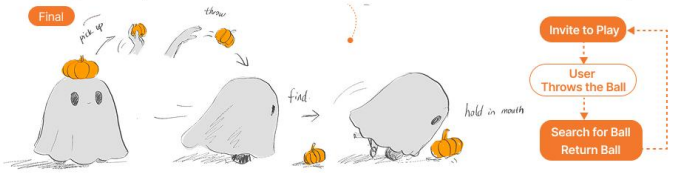
Iteration Process



High feasibility
An interaction similar to playing fetch with a dog

However, once the robot is perceived as a small companion with its **own personality**, throwing its body feels inappropriate; the tossed object should therefore be a **toy** it carries, rather than the robot itself.





Form Design

Iteration Process

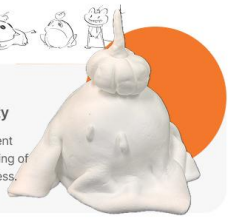


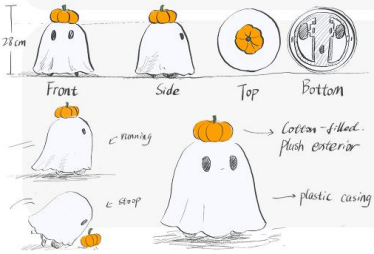
[Why a Ghost?]

Emotional State
Emotional overwhelm often feels closed, isolated, and difficult to express.

Companion Presence
Booji takes the form of a gentle ghost, imagined as a quiet companion.

Floating Quality
Its gliding movement reinforces the feeling of drifting and lightness.





Pumpkin

The pumpkin is sewn in soft fabric and filled with cotton for a soft touch. Magnets in the bottom and sides attach it to the ghost body.

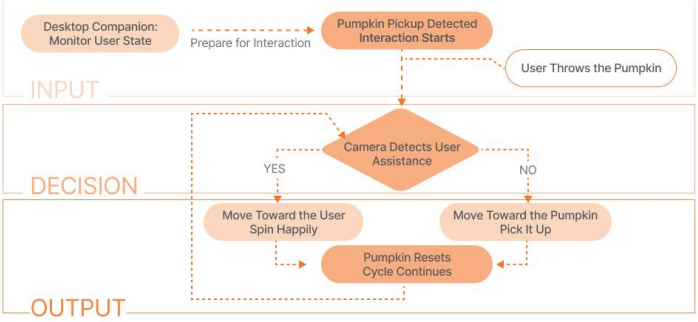
Ghost

The ghost is made of stiffened fabric to hold its shape, balancing a clean appearance with the flexibility needed for later bending movement.

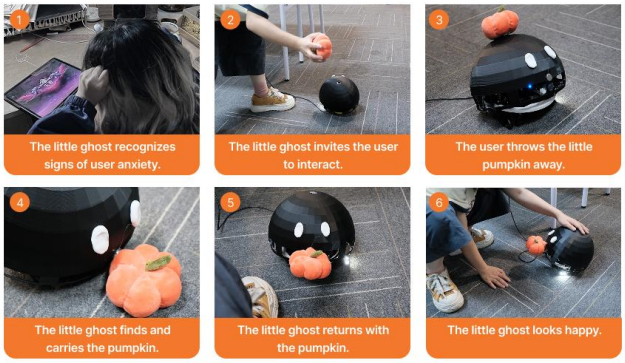
DESIGN ANALYSIS

Interaction Flow

An emotionally light interaction loop designed to encourage playful engagement and responsive companionship.



Storyboard



Structural Design



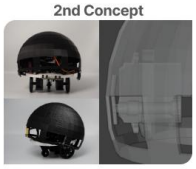
Design Technology Iteration

1st Concept



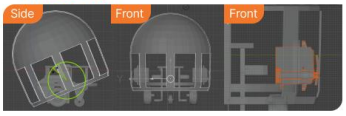
The long connecting rod reduces overall stability.

2nd Concept



Side-mounted servo improves stability, but the low profile weakens the form.

Final Concept



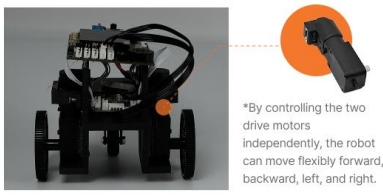
Keeping the side-mounted servo, a taller shell gives the form both stability and visual balance.

The shell tilts forward through two side-mounted servo motors connected by linkages.

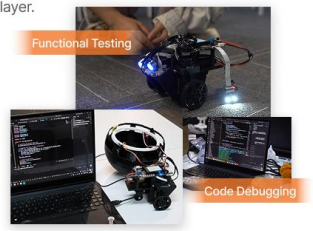
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Main Chassis Structure

Two drive motors are mounted on both sides and connected to the wheels.
The motor driver board sits in the center, with the battery placed at the rear for balance.
The larger main control board is arranged on the second layer.



*By controlling the two drive motors independently, the robot can move flexibly forward, backward, left, and right.



Other Sensors



Infrared Sensor To detect whether the pumpkin has been picked up

Camera Used for searching for pumpkins



Magnet 01
Magnet 02

Used for holding the pumpkin

EXHIBITION DISPLAY



Exhibition Recap







Tutor Feedback

- Replace the fabric with a **hard shell** and increase the overall **height** for a better appearance.
- Add a fetch-like function so the robot can **retrieve the pumpkin** thrown by the user.

At the exhibition, I gave a five-minute presentation of my project and responded to questions from the guest tutors. During the live demonstration, I further explained how the robot works. The on-site discussion provided valuable feedback and helped clarify directions for future improvement.

Future Improvements

- Refine form and structure
- Enhance interactive feedback
- Expand accessories and play modes



Click here to watch the project video.



"Carry the blessings of the torch to more hearts.

02 TORCHLIGHT

Exhibition Design | 2025 / 8 ~ 2025 / 10 ■ Zhang Lie ■ Responsibilities: Preliminary Research, Interaction Flow Design, Exhibition Content Development

Immersion / Interaction / Cultural Learning

- Torch Festival Interactive Exhibition Design, an immersive cultural exhibition inspired by ethnic rituals
- Transforms firelight, patterns, and festival narratives into spatial and interactive elements
- Creates a participatory experience where audiences move, gather, and engage with cultural memory

https://youtu.be/QC7J7yvCPw0

BACKGROUND

The Yi Torch Festival

The Torch Festival is one of the most significant and representative traditional festivals of the Yi people in southwestern China. The festival typically includes the following activities:

FIRE WORSHIP



The fire-worship stage includes offerings, rituals, the lighting of the sacred fire, and a torch procession.

CELEBRATING WITH THE FIRE



The fire-celebration stage includes performances, competitions, beauty contests, and social activities around the sacred fire.

SENDING OFF THE FIRE



The fire-sending stage marks the climax of the Torch Festival, when people carry torches, gather them into great bonfires, and celebrate with singing and dancing.

RESEARCH

Ritual Process | Site Transformation | Collective Participation | Contemporary Evolution

Field Research

Through Tsinghua University's winter field practice in the Greater Liangshan region, I conducted field research on the Yi Torch Festival, focusing on its ritual process, spatial transformation, participation, and cultural context.





Practice Team Poster

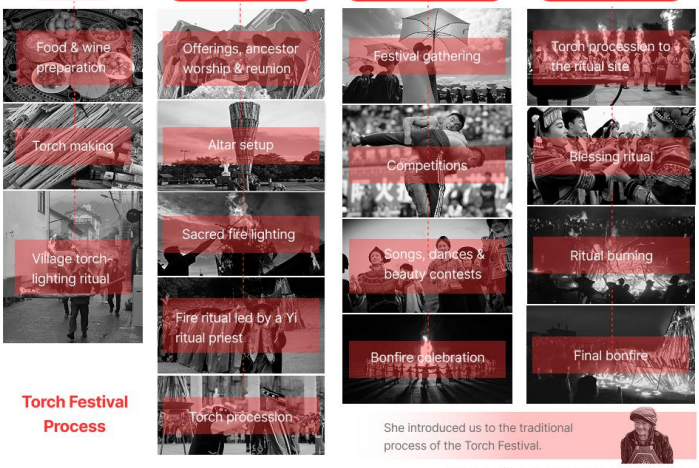


Photos related to practice

What have we done?

- Elder Interviews & Festival Process Documentation → Torch Festival Process
- Site Visits and Interviews with Local Residents → Spatial Transformation
- Village Records Review → Historical and Cultural Context

Preparation → **Day 1** / Fire Worship → **Day 2** / Fire Celebration → **Day 3** / Sending Off



Torch Festival Process

She introduced us to the traditional process of the Torch Festival.

Interview with an elderly villager in Yangpo Village

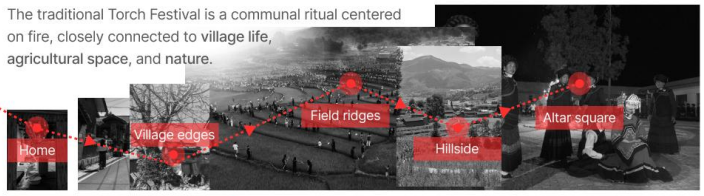
Spatial Transformation

Research Methods

Review of local gazetteers and ethnographic records
On-site visits and observation
Route mapping and spatial comparison

Traditional Village Setting

The traditional Torch Festival is a communal ritual centered on fire, closely connected to village life, agricultural space, and nature.



Contemporary Urban Setting

The urban Torch Festival has shifted from a village-based ethnic ritual to a more public and visible urban celebration.



The core structure of the Torch Festival lies in fire gathering, procession, and collective participation, which forms the basis of this design.

Historical and Cultural Context

Historical Origin

The origin of the Torch Festival has been explained through several myths and legends passed down over time.



1. **Disaster-warding and crop protection:** Heitilaba and Nizhang Azhi led the Yi people to burn away a locust plague with torches, so people later lit torches to honor them.
2. **Resistance against the tusi ruler:** Zhaka led villagers to attack and burn the cruel tusi's home with torches, so people celebrated with fire and dancing.
3. **Uprising against oppression:** The Yi people rose up with torches against the oppressive official Azihudi, and later generations lit torches to remember the victory.

Although these stories vary in form, they often revolve around themes of HEROES DEFEATING EVIL, RESISTING OPPRESSION, and ENDURING LOVE.



- It represents hope that dispels darkness and evil
- A torch that preserves courage, loyalty, and collective memory
- The power to overcome natural disasters and tyranny

Current Representation

What People Usually See



Research Methods

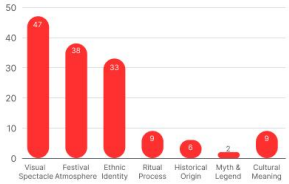
A total of 50 Xiaohongshu posts under the keyword "Torch Festival" were coded through a tag-based content analysis. Each sample could receive multiple tags.

Visual Spectacle | Festival Atmosphere | Myth & Legend | Ethnic Identity | Ritual Process | Cultural Meaning | Historical Origin



statistical graph

Content Representation of the Torch Festival on Xiaohongshu



Analysis

Most Visible	Visual Spectacle / Festival Atmosphere Ethnic Identity
Occasionally Mentioned	Ritual Process / Cultural Meaning / Historical Origin
Nearly Absent	Myth & Legend

The Torch Festival is often presented as spectacle, while its deeper cultural meaning remains less visible.

DESIGN DEFINITION

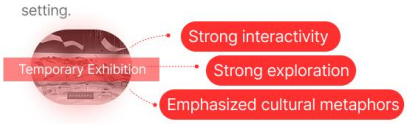
Reconstruction | Cultural Continuity | Playfulness

Research Findings

Cultural Gap	Translate the Torch Festival from visual spectacle into deeper cultural meaning.
Myth Narratives	Extract themes of heroism, resistance, protection, and enduring love.
Site Transformation	Transform fire gathering, torch procession, and collective participation into spatial design logic.
Torch Symbolism	Use the torch to symbolize hope, courage, loyalty, resistance, and memory.

Needs Analysis

As a highly regional festival, the Torch Festival is difficult for most audiences to experience in person. This project translates its culture and history into an accessible special exhibition in a museum setting.



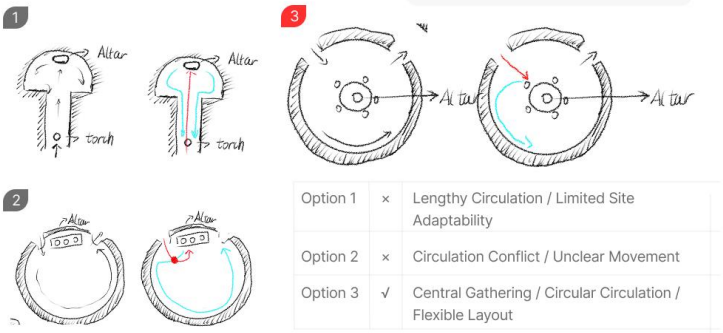
Design Requirements
Engaging Visuals | Modular Structure | Interactive Experience

Spatial Structure

Field research shows that the Torch Festival contains both "gathering" and "dispersal" spatial patterns. "Gathering" is reflected in the central assembly around the torch, while "dispersal" appears through processions, movement, and flowing paths.

Spatial Design Sketches

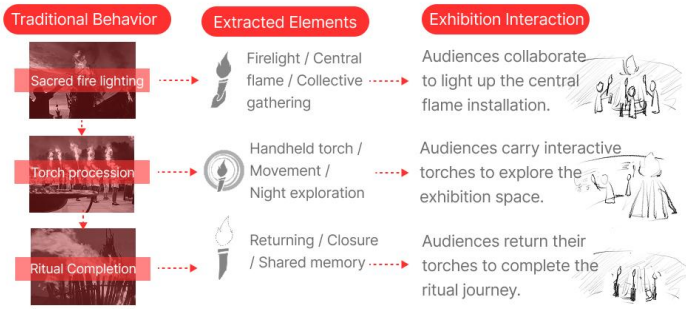
Design Strategy:
Central Gathering + Circular Circulation



Option 1	×	Lengthy Circulation / Limited Site Adaptability
Option 2	×	Circulation Conflict / Unclear Movement
Option 3	√	Central Gathering / Circular Circulation / Flexible Layout

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Ritual Behavior Translation



Exhibition Element Extraction

The design extracts firelight, night atmosphere, collective participation, and ritual interaction from the Torch Festival. To meet museum safety requirements, real flames are translated into lighting, projection, and interactive installations.

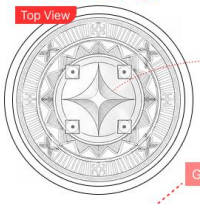
PROTOTYPE SHOWCASE



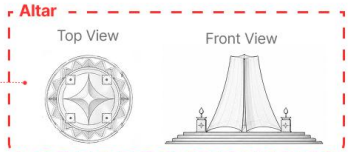
Exhibition Space Design

The altar design is inspired by traditional Yi sun patterns. Its concentric circular structure and radial geometric motifs create a sense of ritual order, while the central gauze installation symbolizes rising flames and serves as a projection surface for firelight and cultural content.

Altar Area Design

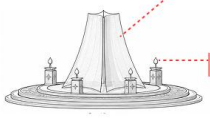


Altar





traditional Yi sun patterns



Gauze Projection Projectors hidden above or behind the installation cast firelight and dynamic visuals onto the translucent scrim, creating a floating flame effect without using real fire.

Torch An infrared sensor at the base detects when the torch is lifted and triggers the interactive response.



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Torch Design



The upward-tapering form symbolizes rising flames, unity, and the connection between humans and the spiritual world.



Fire-related Yi motifs were extracted from pattern research and transformed into a decorative flame pattern for the torch design.



100 Yi patterns were analyzed, and 20 representative motifs were selected.

- Black as the dominant color, with red and yellow as accents
- Motifs include plants, animals, geometry, and human figures
- Frequent use of spiral curves



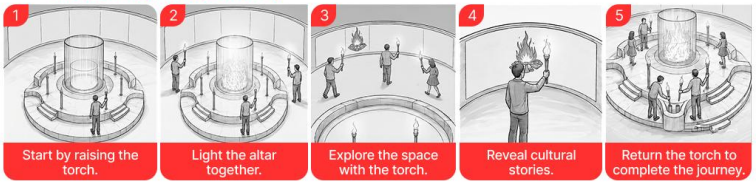
The upward spiral structure symbolizes the accumulation of energy and the eruption of flames.



DESIGN ANALYSIS

Interaction Flow

Audience Experience Journey



This interaction emphasizes collective participation, echoing the communal nature of the Torch Festival.

Interaction System Framework

	Input	Logic	Output
Altar System	Torch lift detection	At least two torches lifted	Altar lighting animation
LED System	Light sensor threshold	Torch light reaches the trigger point	Cultural animation

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Click here to watch the project video.



03 WOVEN DISCIPLINES

Data Visualization | 2025 / 8 ~ 2025 / 10 ■ Wu Qiong ■ Responsibilities: Concept Design, Data Organization & Analysis based on Tsinghua University and Aalto University

Information Visualization | Interdisciplinary Design | Data Storytelling

- **An information visualization project exploring interdisciplinary connections in design.**
- Based on data from five leading design schools, it analyzes courses, research directions, and disciplinary links.
- Inspired by dreamcatchers, it transforms complex relationships into a woven visual map of design knowledge.

BACKGROUND

Design | Technology | Society

COMPLEX CHALLENGES	DESIGN AS A BRIDGE	INSTITUTIONAL DIFFERENCES
Social, technological, and cultural issues are increasingly connected, making single-discipline solutions insufficient.	Design education is increasingly positioned between technology, art, and society, promoting interdisciplinary teaching.	Leading design schools, including Aalto University, RCA, Parsons, and Tsinghua University, have developed interdisciplinary practices with different structures, methods, and priorities.

Systematic comparative research on these interdisciplinary curriculum models remains limited.

RESEARCH

[Interdisciplinary Courses in Leading Design Schools Worldwide]

Research Questions

- How Other Disciplines Intersect with Design and Art
- Differences in the Structure, Scope, and Focus of Interdisciplinary Courses Across Institutions
- The Roles Design Plays in Interdisciplinary Curriculum Systems

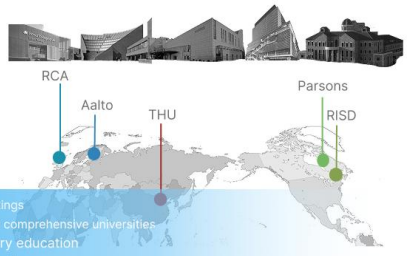
Research Objectives

1. **Map Course Distribution** — Compare the scale and proportion of interdisciplinary courses across schools.
2. **Identify Structural Patterns** — Analyze disciplinary connections, core nodes, and network tendencies.
3. **Compare Institutional Models** — Examine differences in curriculum focus, scope, and organization.
4. **Reflect on Educational Logic** — Discuss whether interdisciplinarity supplements, integrates, or reshapes design education.

Scope & Data

Scope

Royal College of Art
Parsons School of Design
Rhode Island School of Design
Aalto University
Tsinghua University



Selection Criteria

- **Academic Influence** — QS Art & Design rankings
- **Institutional Diversity** — design schools and comprehensive universities
- **Interdisciplinary Focus** — cross-disciplinary education
- **Data Accessibility** — public course information

Data Sources

Public information from official institutional websites, including:

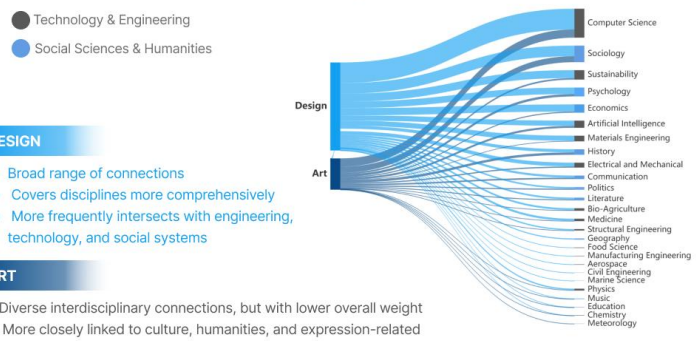
- Official course pages
- Program and department descriptions
- Project and course descriptions

Methodology

Step 1 | Course Collection	Collected public course information from five institutions.
Step 2 Course Classification	Classified courses into design, art, interdisciplinary, and practice-based categories.
Step 3 Discipline Tag Standardization	Grouped course-related fields into consistent disciplinary tags such as technology, society, business, sustainability, AI, media, culture, engineering, and art.
Step 4 Visualization Analysis	Translated the data into chord diagrams, radar charts, word clouds, and proportion charts to compare interdisciplinary patterns across schools.

Overall Findings

Overall Course Intersections Across Five Design Schools



DESIGN

- Broad range of connections
- Covers disciplines more comprehensively
- More frequently intersects with engineering, technology, and social systems

ART

- Diverse interdisciplinary connections, but with lower overall weight
- More closely linked to culture, humanities, and expression-related fields
- Plays a complementary role in perception, critique, and expression within interdisciplinary education

School-Specific Analysis

Aalto University
School of Arts, Design and Architecture

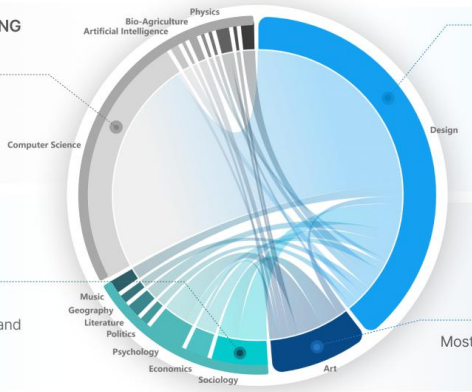
Feature: Computer Science as the Core Technology Anchor
Computer science occupies a prominent share and connects closely with design, art, and sociology, highlighting its foundational role in interdisciplinary courses.

TECHNOLOGY & ENGINEERING

Computer Science
Most frequent interdisciplinary connections in Technology and Engineering
Share: **79.1%**

SOCIAL SCIENCES & HUMANITIES

Sociology
Most frequent interdisciplinary connections in Social Sciences and Humanities
Share: **34.29%**



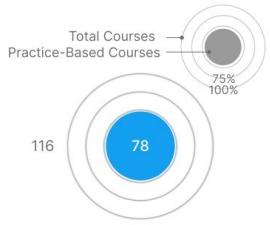
DESIGN & APPLICATION

Computer Science
Highest share of interdisciplinarity
Share: **15.6%**

DESIGN & APPLICATION

Sociology
Most frequent interdisciplinary connections
Share: **30%**

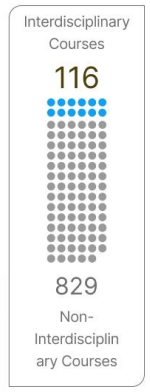
Interdisciplinary Status of Undergraduate and Graduate Courses



Practice-Based Course Share



30 Project Keywords



UG & Graduate Interdisciplinary Courses

清華大學美術學院
Academy of Arts & design, Tsinghua University

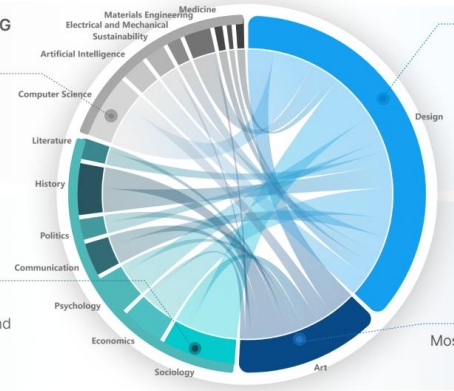
Feature: Art–Technology Integration with Strategic and Industrial Focus
While the undergraduate curriculum leans toward the social sciences and humanities, research projects focus more on technology, especially artificial intelligence, human–computer interaction, and cultural heritage.

TECHNOLOGY & ENGINEERING

Computer Science
Most frequent interdisciplinary connections in Technology and Engineering
Share: **36.2%**

SOCIAL SCIENCES & HUMANITIES

Sociology
Most frequent interdisciplinary connections in Social Sciences and Humanities
Share: **25.8%**



DESIGN & APPLICATION

Sociology
Highest share of interdisciplinarity
Share: **14.4%**

DESIGN & APPLICATION

History
Most frequent interdisciplinary connections
Share: **21.7%**

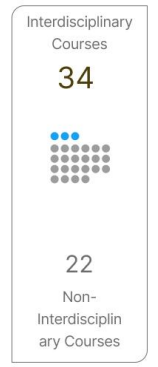
Interdisciplinary Status of Undergraduate and Graduate Courses



Practice-Based Course Share



30 Project Keywords



UG & Graduate Interdisciplinary Courses

Royal College of Art

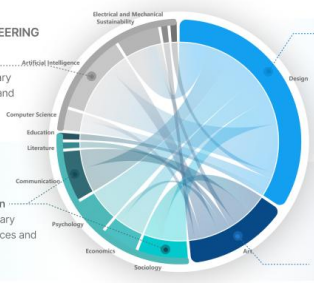
30 Project Keywords

TECHNOLOGY & ENGINEERING

Artificial Intelligence
Most frequent interdisciplinary connections in Technology and Engineering
Share: **50%**

SOCIAL SCIENCES & HUMANITIES

Sociology & Communication
Most frequent interdisciplinary connections in Social Sciences and Humanities
Share: **26.9%**

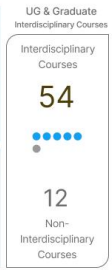


DESIGN & APPLICATION

Artificial Intelligence
Highest share of interdisciplinarity
Share: **16.3%**

DESIGN & APPLICATION

Sociology & Communication & Artificial Intelligence
Most frequent interdisciplinary connections
Share: **17.4%**



UG & Graduate Interdisciplinary Courses

Interdisciplinary Status of Undergraduate and Graduate Courses

Feature: Research-Driven and Collaborative Across the School
Compared with other schools, the program places greater emphasis on research projects. Required interdisciplinary units and a dual-degree program with Imperial College London further strengthen collaboration across disciplines, addressing complex global challenges such as sustainability and healthcare.

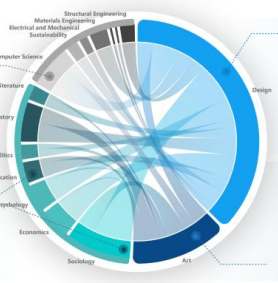
THE NEW SCHOOL PARSONS

TECHNOLOGY & ENGINEERING

Computer Science
Most frequent interdisciplinary connections in Technology and Engineering
Share: **41.2%**

SOCIAL SCIENCES & HUMANITIES

Sociology
Most frequent interdisciplinary connections in Social Sciences and Humanities
Share: **26.9%**

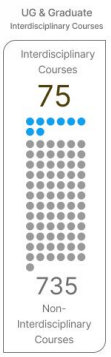


DESIGN & APPLICATION

Sociology
Highest share of interdisciplinarity
Share: **16.3%**

DESIGN & APPLICATION

Sociology
Most frequent interdisciplinary connections
Share: **23.3%**



UG & Graduate Interdisciplinary Courses

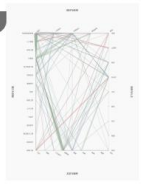
Interdisciplinary Status of Undergraduate and Graduate Courses

Feature: Research-Driven and Collaborative Across the School
Parsons focuses more on intersections between design and the humanities, such as design with sociology or history. Although it includes some technology-related courses, such as design with computer science, its overall emphasis is on social issues rather than technological frontiers.

DESIGN CONCEPT

Design | Technology | Society

Design Association



Idea 1: Carpet



Idea 2: Age of Exploration



Idea 3: Dreamcatcher

[Why a Dreamcatcher?]

Aspirational Symbol

A dreamcatcher represents catching dreams and protecting hope, echoing the goal of helping students pursue their aspirations.

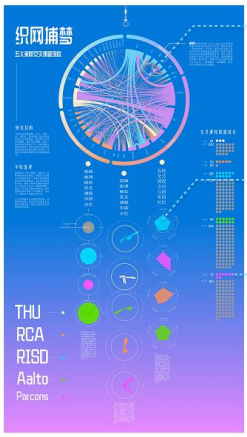
Connected Structure

Its circular web visualizes the links between disciplines, turning interdisciplinary relationships into a clear and poetic form.

Data Extension

The hanging elements can carry extra icons and data, allowing the visualization to expand from multiple perspectives.

DESIGN SHOWCASE



Main visual:

A chord diagram is used to show the intersections between the design discipline and other fields, helping viewers quickly form an initial impression of interdisciplinary connections.

Supporting Charts

Three chart types compare each school's interdisciplinary courses across practice, disciplinary intersections, and Tsinghua's course directions.

Dot plots visually show the number of courses and the proportion of interdisciplinary courses.

